

Principal Component Analysis: From Classical Ideas to Geometric and Statistical Analysis

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Why PCA? (1/2)

- Modern datasets: $X_1, \dots, X_n \in \mathbb{R}^d$ with
 - d large (tens, hundreds, thousands).
 - Strong correlations between coordinates.
- Problems:
 - Visualization is impossible in high dimensions.
 - Learning algorithms may overfit or become unstable.
 - Computation and storage become expensive.
- Idea: replace the original variables by a smaller set of *linear combinations* that capture most of the structure.

Why PCA? (2/2)

- PCA is an unsupervised method.
- No labels Y ; we use only the geometry of X (through covariance).
- Typical goals:
 - Data compression / dimensionality reduction.
 - Visualization in 2D/3D.
 - Denoising.
 - As a preprocessing step before regression / classification.
- For MSc/PhD level:
 - PCA is also a key example of spectral methods.
 - It is a clean model to study statistical risk and geometry.

Toy Geometric Picture (2D/3D)

- Imagine points in $\mathbb{R}^2$ forming an elongated cloud.
- First principal component (PC1):
 - A unit vector u_1 along the longest direction of variability.
 - Projections $\langle u_1, X \rangle$ have maximal variance.
- Second principal component (PC2):
 - A unit vector u_2 orthogonal to u_1 .
 - Captures the next largest variance, orthogonal to PC1.
- In $\mathbb{R}^3$, PCA often finds a plane where the cloud lies approximately.

Key intuition: PCA finds a low-dimensional *linear* subspace that best fits the data in a mean squared error sense.

Two Classical Viewpoints of PCA

① **Max-variance view:**

- First PC: direction of maximal variance.
- Next PCs: directions of maximal variance orthogonal to previous ones.

② **Min-error view:**

- Find a k -dimensional subspace such that orthogonal projection onto that subspace minimizes the mean squared reconstruction error.

These views are equivalent and both will be useful:

- For implementation and interpretation: max-variance/SVD view.
- For theory and geometry: reconstruction-error view.

Centering and Covariance

- Observations: $X_1, \dots, X_n \in \mathbb{R}^d$.
- Empirical mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

- Centered observations:

$$\tilde{X}_i = X_i - \bar{X}.$$

- Empirical covariance:

$$\Sigma_n = \frac{1}{n} \sum_{i=1}^n \tilde{X}_i \tilde{X}_i^\top \in \mathbb{R}^{d \times d}.$$

- PCA is typically applied to $\tilde{X}_i$ and Σ_n .

Eigen-Decomposition of Σ_n

- Σ_n is symmetric positive semidefinite.
- It has eigen-decomposition

$$\Sigma_n = V\Lambda V^\top,$$

where

$$V = [v_1, \dots, v_d], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d),$$

and

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0.$$

- v_j are called *principal directions* (or loadings).
- λ_j are variances along these directions.

First Principal Component as a Rayleigh Quotient

- Let $u \in \mathbb{R}^d$ with $\|u\|_2 = 1$.
- Variance of projection:

$$\text{Var}(\langle u, X \rangle) \approx u^\top \Sigma_n u.$$

- First principal component solves:

$$u_1 \in \arg \max_{\|u\|_2=1} u^\top \Sigma_n u.$$

- By Rayleigh quotient theory:

$$u_1 = v_1, \quad \text{the eigenvector with largest eigenvalue } \lambda_1.$$

- Next components:

$$u_k \in \arg \max_{\|u\|_2=1, u \perp u_1, \dots, u_{k-1}} u^\top \Sigma_n u,$$

giving $u_k = v_k$.

Optimization View: Reconstruction Error

- Let $U \in \mathbb{R}^{d \times k}$ with $U^\top U = I_k$.
- Projection of $\tilde{X}_i$ onto the subspace spanned by columns of U :

$$P_U(\tilde{X}_i) = UU^\top \tilde{X}_i.$$

- Reconstruction error for point i :

$$\left\| \tilde{X}_i - UU^\top \tilde{X}_i \right\|_2^2.$$

- Empirical risk:

$$\tilde{R}_n(U) := \frac{1}{2n} \sum_{i=1}^n \left\| \tilde{X}_i - UU^\top \tilde{X}_i \right\|_2^2.$$

- PCA subspace of dimension k :

$$U_n = [v_1, \dots, v_k] \in \arg \min_{U^\top U = I_k} \tilde{R}_n(U).$$

Quick Derivation: Minimizing Reconstruction Error

Sketch:

$$\begin{aligned}\tilde{R}_n(U) &= \frac{1}{2n} \sum_{i=1}^n \|\tilde{X}_i - UU^\top \tilde{X}_i\|_2^2 \\&= \frac{1}{2n} \sum_{i=1}^n \left(\|\tilde{X}_i\|_2^2 - \|U^\top \tilde{X}_i\|_2^2 \right) \\&= \frac{1}{2n} \sum_{i=1}^n \|\tilde{X}_i\|_2^2 - \frac{1}{2n} \sum_{i=1}^n \tilde{X}_i^\top UU^\top \tilde{X}_i \\&= \text{constant} - \frac{1}{2} \text{Tr}(U^\top \Sigma_n U).\end{aligned}$$

- So minimizing $\tilde{R}_n(U)$ is the same as maximizing $\text{Tr}(U^\top \Sigma_n U)$.
- The maximizer is $U_n = [v_1, \dots, v_k]$ by a “block Rayleigh quotient” argument.
- This expression (risk = constant – block Rayleigh quotient) is key for the geometric analysis later.

SVD View and Scores

- Data matrix $X \in \mathbb{R}^{n \times d}$ with rows $\tilde{X}_i^\top$.
- SVD:

$$X = WSV^\top,$$

with $W \in \mathbb{R}^{n \times d}$, $S = \text{diag}(s_1, \dots, s_d)$, $V = [v_1, \dots, v_d]$.

- Then

$$\Sigma_n = \frac{1}{n} X^\top X = V \frac{S^2}{n} V^\top.$$

- Principal component scores:

$$Z = XV_k = WS_k, \quad V_k = [v_1, \dots, v_k].$$

Explained Variance and Choosing k

- Total variance:

$$\text{tr}(\Sigma_n) = \sum_{j=1}^d \lambda_j.$$

- Variance explained by first k PCs:

$$\sum_{j=1}^k \lambda_j.$$

- Proportion of variance explained (PVE):

$$\text{PVE}(k) = \frac{\sum_{j=1}^k \lambda_j}{\sum_{j=1}^d \lambda_j}.$$

- Practical heuristics:

- Scree plot (look for “elbow”).
- Choose smallest k with $\text{PVE}(k) \geq$ (e.g.) 0.9.
- Application-specific trade-off: model complexity vs information retained.

Suggestion for 90-minute slot:

- On the board, work through a tiny numerical example:
 - $d = 2$, $n = 3$ or 4 .
 - Compute Σ_n , eigenvalues, eigenvectors.
 - Show projections and reconstruction.
- Ask students:
 - What happens if we do not center the data?
 - When might PCA be a bad idea (nonlinear structure, heavy tails, etc.)?

- Random vector $X \in \mathbb{R}^d$ with mean $m = \mathbb{E}[X]$ and covariance

$$\Sigma = \mathbb{E}[(X - m)(X - m)^\top].$$

- Eigen-decomposition:

$$\Sigma = U\Lambda U^\top, \quad U = [u_1, \dots, u_d], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d).$$

- The population k -dimensional PCA subspace:

$$U^* = [u_1, \dots, u_k].$$

- Sample PCA uses $U_n = [v_1, \dots, v_k]$ based on Σ_n as an estimator of U^* .

Population Reconstruction Risk

- For any $U \in \mathbb{R}^{d \times k}$ with $U^\top U = I_k$, define the population risk:

$$R(U) := \frac{1}{2} \mathbb{E} \left[\left\| X - \mathbb{E}[X] - UU^\top (X - \mathbb{E}[X]) \right\|_2^2 \right].$$

- One can show:

$$R(U) = \frac{1}{2} \text{tr}(\Sigma) - \frac{1}{2} \text{tr}(U^\top \Sigma U).$$

- Therefore, $U^* = [u_1, \dots, u_k]$ minimizes $R(U)$.
- Now we start to think of PCA as choosing a *parameter* (a subspace) to minimize a risk functional.

- The risk $R(U)$ depends only on the subspace $\mathcal{S} = \text{span}(U)$: if $V = UQ$ with Q orthogonal, then

$$VV^{\top} = UU^{\top}, \quad R(V) = R(U).$$

- We want a space where each point is a k -dimensional subspace:
 - Stiefel manifold: orthonormal frames.
 - Grassmann manifold: subspaces (frames modulo rotations).

Stiefel Manifold

$$\text{St}(d, k) := \{U \in \mathbb{R}^{d \times k} : U^\top U = I_k\}.$$

Grassmann Manifold

$$\text{Gr}(d, k) := \text{St}(d, k) / \sim,$$

where $U \sim V$ if $V = UQ$ for some orthogonal $Q \in \mathbb{R}^{k \times k}$.

- Each $[U] \in \text{Gr}(d, k)$ is a k -dimensional subspace of $\mathbb{R}^d$.
- PCA:

$$[U^*] \in \arg \min_{[U] \in \text{Gr}(d, k)} R([U]), \quad [U_n] \in \arg \min_{[U] \in \text{Gr}(d, k)} R_n([U]).$$

Geometry of $\text{Gr}(d, k)$ (Sketch)

- Tangent space at $[U]$:

$$T_{[U]}\text{Gr}(d, k) \simeq \{\Delta \in \mathbb{R}^{d \times k} : U^\top \Delta = 0\}.$$

- Principal angles $\theta_1, \dots, \theta_k$ between $[U]$ and $[V]$:
 - Compute SVD of $U^\top V$.
 - Cosines of the angles are the singular values.
- Riemannian distance:

$$\text{dist}^2([U], [V]) = \sum_{j=1}^k \theta_j^2.$$

- For statistical analysis:
 - Use exponential map $\exp_{[U]}$ to move along geodesics.
 - Use logarithm map $\text{Log}_{[U]}$ to map nearby subspaces to the tangent space.

Two Notions of Error

- **Geometric error:**

$$\text{dist}([U_n], [U^*]) \quad (\text{distance on } \text{Gr}(d, k)).$$

- **Excess risk:**

$$E_n = R([U_n]) - R([U^*]).$$

- Both are random (depend on the sample).
- We are interested in:
 - Convergence: $[U_n] \rightarrow [U^*]$ as $n \rightarrow \infty$.
 - Rate and limiting distribution (CLT-type results).
 - How E_n behaves and what determines it.

Theorem (Informal): Asymptotic Normality on $\text{Gr}(d, k)$

Under assumptions such as:

- Eigengap: $\lambda_k > \lambda_{k+1}$.
- Finite moments (up to order 4) of projections $\langle u_j, X \rangle$.

one can show:

- **Consistency:**

$$\text{dist}([U_n], [U^*]) \xrightarrow{P} 0.$$

- **Asymptotic normality:**

$$\sqrt{n} \text{Log}_{[U^*]}([U_n]) \xrightarrow{d} G,$$

where G is a Gaussian matrix in the tangent space $T_{[U^*]}\text{Gr}(d, k)$ with explicitly described covariance.

- **Excess risk:**

$$nE_n \xrightarrow{d} \frac{1}{2} \|H\|_F^2,$$

where H is another Gaussian matrix with covariance linked to G and the curvature of the risk.

Interpretation of the Asymptotic Result

- This is an analogue of the CLT for estimators:
 - In Euclidean settings, $\sqrt{n}(\hat{\theta}_n - \theta^*)$ converges to a Gaussian.
 - Here $\hat{\theta}_n$ is a point on $\text{Gr}(d, k)$.
 - We compare it to $\theta^* = [U^*]$ via the log map into the tangent space.
- The distribution of G and H depends on:
 - Eigengaps $\lambda_j - \lambda_{k+i}$.
 - Fourth-order moments $\mathbb{E}[\langle u_{k+i}, X \rangle^2 \langle u_j, X \rangle^2]$.
- So PCA risk is not determined solely by Σ ; higher moments matter.

- For $\delta \in (0, 1)$ define the $(1 - \delta)$ -quantile of E_n :

$$Q_{E_n}(1 - \delta) := \inf\{t \in \mathbb{R} : \mathbb{P}(E_n \leq t) \geq 1 - \delta\}.$$

- As $n \rightarrow \infty$, $nQ_{E_n}(1 - \delta)$ converges to a quantity depending on:
 - Mixed moments $\mathbb{E}[\langle u_{k+i}, X \rangle^2 \langle u_j, X \rangle^2]$.
 - Eigengaps $\lambda_j - \lambda_{k+i}$.
- This gives distribution-aware asymptotic confidence bands for PCA excess risk.

Block Rayleigh Quotient

- Recall:

$$R([U]) = \frac{1}{2}\mathbb{E}[\|X\|_2^2] - \frac{1}{2}\text{Tr}(U^\top \Sigma U).$$

- Define

$$F([U]) = -\frac{1}{2}\text{Tr}(U^\top AU), \quad A = \Sigma.$$

- This is the *block Rayleigh quotient*.
- $[U^*]$ is a minimizer of F (equivalently of R).
- To control non-asymptotic behavior, we study the curvature of F on $\text{Gr}(d, k)$ near $[U^*]$.

Self-Concordance Along Geodesics (Idea)

- Consider a geodesic $\gamma(t)$ between $[U^*]$ and another point $[U]$.
- Study the 1D function $g(t) = F(\gamma(t))$.
- Proposition (informal):
 - If the maximum principal angle between $[U]$ and $[U^*]$ is $< \pi/4$, then g satisfies a generalized self-concordance condition of the form

$$|g'''(t)| \leq C(\theta) g''(t)$$

along the geodesic.

- Consequence:
 - Second-order Taylor expansions of F around $[U^*]$ are well controlled within a neighborhood (no wild changes in the Hessian).
 - This is analogous to Bach's self-concordant analysis of logistic regression, but now on a manifold.

Non-Asymptotic Excess Risk (High-Level Statement)

Under eigengap and moment assumptions, and for n above a certain threshold, one can prove that, with high probability $1 - \delta$,

$$E_n = R([U_n]) - R([U^*]) \leq \frac{C}{n} \sum_{i=1}^{d-k} \sum_{j=1}^k \frac{\mathbb{E}[\langle u_{k+i}, X \rangle^2 \langle u_j, X \rangle^2]}{\lambda_j - \lambda_{k+i}},$$

where

- C depends on δ and constants in the self-concordance and concentration arguments.
- The right-hand side matches (up to constants) the asymptotic limit.

Interpretation:

- For large enough n , PCA's excess risk decays as $1/n$.
- The leading constant encodes both spectral structure and higher moments.

Spiked Covariance Model

- A simple but important model:

$$X = Z + \varepsilon,$$

where

- Z lives in a k -dimensional subspace (low-rank signal),
 - $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_d)$ (isotropic noise),
 - Z and ε are independent.
- Then covariance

$$\Sigma = S + \sigma^2 I_d,$$

where S has rank k and encodes the signal.

- The top- k eigenvectors of Σ recover the signal subspace.

Asymptotic Behavior in the Spiked Model

- In this model, the Gaussian matrices appearing in the asymptotic characterization of PCA simplify.
- The variances of entries of the limiting Gaussian G and H can be written in closed form in terms of:
 - noise variance σ^2 ,
 - signal eigenvalues (non-zero eigenvalues of S).
- This yields explicit formulas for the asymptotic distribution of:
 - geometric error $\text{dist}([U_n], [U^*])$,
 - scaled excess risk nE_n .
- Good sandbox model to connect theory and simulation.

- Simulate the spiked covariance model:
 - Fix d, k, σ^2 , and signal eigenvalues.
 - Generate $X_1, \dots, X_n$ and compute sample PCA.
- Empirically estimate:
 - distribution of $\text{dist}([U_n], [U^*])$,
 - distribution of nE_n .
- Compare with asymptotic predictions from the theory.
- Explore the impact of:
 - decreasing eigengap,
 - increasing noise σ^2 ,
 - heavier tails for Z (to see role of fourth moments).

Summary of the Lecture

- Classical PCA:
 - Eigen-decomposition / SVD framework.
 - Equivalent max-variance and min-error formulations.
- Geometric viewpoint:
 - Parameter is a subspace $\Rightarrow$ point on $\text{Gr}(d, k)$.
 - Use distances and tangent spaces to measure errors.
- Statistical performance:
 - PCA behaves as an M -estimator on a manifold.
 - Asymptotic normality of subspace error.
 - Excess risk E_n decays like $1/n$ with a distribution-aware constant.
- Tools:
 - Principal angles, Grassmannian geometry.
 - Block Rayleigh quotient and generalized self-concordance.

Classical PCA and spectral methods

- I.T. Jolliffe. *Principal Component Analysis*. Springer.
- T. Hastie, R. Tibshirani, J. Friedman. *The Elements of Statistical Learning*, Ch. 14.

Geometric and statistical analysis of PCA

- A. El Hanchi, M. A. Erdogdu, C. J. Maddison. *A Geometric Analysis of PCA*.

Thank You

Questions?